Beacon's Light College, Kandhkot

Quiz 1 – Mathematics

Chapter 1: Complex Numbers

Student Name: ________________________
Class: First Year
Subject: Mathematics

MULTIPLE CHOICE QUESTIONS

1) Modulus of a+ib is: (a) a^2+b^2 (b) $\sqrt{(a^2+b^2)}$ (c) a^2-b^2 (d) a–ib
2) p/q, $p,q \in Z \wedge q \neq 0$ is set of all: (a) Natural (b) Integers (c) Irrational (d) Rational
3) $|2-3i|$ = (a) 2 (b) 3 (c) $\sqrt{13}$ (d) $\sqrt{17}$
4) $9a^2+64b^2$= (a) 3a+8b (b) (3a+8b)(3a–8b) (c) (3a+8b)(3a–8b) (d) None
5) $|a+ib|$ is equal to: (a) $\sqrt{(a^2+b^2)}$ (b) a^2-b^2 (c) $-a^2-b^2$ (d) None
6) The property of inequality used in $a>b$ and $b>c \Rightarrow a>c$ is: (a) Additive (b) Multiplicative (c) Transitive (d) Additive inverse
7) Which set has closure property with respect to addition: (a) {–1,1} (b) {–1} (c) {1} (d) {0}
8) $a,b \in R, a=b \Rightarrow b=a$ is called: (a) Reflexive (b) Symmetric (c) Multiplicative (d) None
9) i^2 = (a) i (b) 1 (c) –1 (d) None
10) If n is a prime number then $\sqrt{n}$ is: (a) Irrational (b) Rational (c) Prime (d) None
11) For all $a,b,c \in R$, $c>0$, $a>b \Rightarrow ac>bc$ is: (a) Reflexive (b) Symmetric (c) Additive (d) Multiplicative
12) 1/7 is called: (a) Rational (b) Irrational (c) Integer (d) None
13) $9a^2+16b^2$= (a) (3a+4b)(3a–4b) (b) (3a+4b)(3a–4b) (c) (3a+4b)(3a+4b) (d) None
14) Geometrically the modulus of a complex number represents its distance from: (a) (1,0) (b) (0,1) (c) (0,0) (d) (1,1)
15) i^{1}■ = (a) –1 (b) 1 (c) i (d) –i
16) $(-i)^{212}$ = (a) –i (b) i (c) 1 (d) –1
17) The number 1 is a: (a) Prime (b) Irrational (c) Even (d) Odd
18) $\sqrt{3}$ is: (a) Odd (b) Natural (c) Rational (d) Irrational
19) Multiplicative property of order of real numbers: (a) $a>b \wedge c>0 \Rightarrow ac>bc$ (b) $a>b \wedge c>0 \Rightarrow acb \wedge c>0 \Rightarrow ac=bc$ (d) $a>b \wedge c<0 \Rightarrow ac>bc$
20) –a–ib is equal to: (a) a+ib (b) –a+ib (c) a–ib (d) –a–ib

WORD PROBLEMS (Scenario-Based)

1) A ship moves 4 km east and then 3 km north. Represent its total displacement as a complex number and find its modulus.
2) An electric current in a circuit is represented by the complex number 5+2i. Find the magnitude of this current.
3) A particle moves from point A(0,0) to B(6,8). Represent the movement using a complex number and find its distance from the origin.